

BNU Computer Networks Project – Phase 2 Work Allocation

MEMBER 1 – MUHAMMAD AREZ Role:(let me know if you want this role, it is essentially making the backbone to which all of the other tasks will connect to)

Integration Lead & Core Infrastructure Estimated Workload: 35–40%

Build:

Core Infrastructure

- R-01 Router
- CS-01 Core Switch
- SW-SRV
- ISP Connection

Server Farm

- SRV-01 (DHCP + DNS)
- SRV-02 (LMS/Web Server)
- SRV-03 (Email Server)
- SRV-04 (Backup Server) Configure:
- VLAN Creation
- VLAN 10 • VLAN 20 • VLAN 40 • VLAN 50 • VLAN 60
- VLAN 70
- Router-on-a-Stick
- Router Subinterfaces
- DHCP Relay (ip helper-address)
- NAT/PAT
- DNS Records
- DHCP Pools
- Server IP Assignments
- Trunk Links
- Internet Connectivity Integration Duties:
- Import all departmental topologies
- Connect departments to CS-01
- Verify VLAN membership
- Perform final testing
- Troubleshoot integration issues

Deliverables: Core topology complete Server farm complete All services functional Final integrated .pkt file

MEMBER 2 – SHEIKH MUHAMMAD IBRAHIM

Role: Admission & Exam Department Developer Estimated Workload: 20–25%

Admission Department (VLAN 10)

Build:

- SW-ADM
- ADM-PC01 to ADM-PC12
- ADM-LT01
- ADM-SRV01 Configure:
- Correct hostnames
- Device placement
- Static IP for ADM-SRV01
- All other devices set to DHCP

Exam Department (VLAN 20)

Build:

- SW-EXAM
- EXAM-PC01 to EXAM-PC15
- EXAM-SRV01
- EXAM-SRV02 Configure:
- Correct hostnames
- Device placement
- Servers configured according to report
- End devices set to DHCP

Deliverables: Complete Admission topology Complete Exam topology Cleanly labelled devices

MEMBER 3 – SAAD MUGHAL Role: Admin &
Finance Department Developer Estimated Workload: 20–25%

Admin Department (VLAN 40)

Build:

- SW-ADMIN
- ADMIN-PC01 to ADMIN-PC15
- ADMIN-SRV01 Configure:
 - Correct hostnames
 - Static IP for ADMIN-SRV01
 - DHCP for all PCs Finance Department (VLAN 50)

Build:

- SW-FIN
- FIN-PC01 to FIN-PC13 Configure:
 - Correct hostnames
 - DHCP for all PCs

Deliverables: Complete Admin topology Complete Finance topology Correct naming convention

MEMBER 4 – MUHAMMAD ISMAIL RANA Role:

Marketing Developer & Testing Coordinator Estimated Workload: 20–25%

Marketing Department (VLAN 60)

Build:

- SW-MKT
- MKT-PC01 to MKT-PC08
- MKT-LT01 to MKT-LT04
- AP-MKT Configure:
 - AP Hostname
 - AP Static IP (10.10.6.200)
 - SSID: BNU-Marketing
 - WEP Security
 - DHCP for laptops and PCs

Testing Responsibilities After

integration:

Perform and document:

- DHCP tests
- DNS tests
- Inter-VLAN connectivity tests
- LMS accessibility tests
- Internet access tests
- Wireless connectivity tests

Collect screenshots for submission.

Deliverables: Complete Marketing topology Functional wireless network Testing screenshots Verification report

FINAL INTEGRATION RULE REMINDER:

Department Members SHALL NOT create:

- R-01
- CS-01
- SW-SRV
- ISP devices
- Router subinterfaces
- NAT configuration
- DHCP Relay
- DNS services
- VLAN trunks to the core

Only one member will perform final integration, who ever is developing the core.